

# §4022(c) Allocation

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# Recoveries

PBGC asserts claims for Due and Unpaid Employer Contributions ("DUEC"), Unfunded Benefit Liabilities ("UBL"), and unpaid premiums against the plan sponsor.

The total value of PBGC's recoveries is allocated to

- UBL claim
  - the payment of UNGB under ERISA §4022(c)
  - SPARR
- DUEC claim
  - increases plan assets
  - SPDRR

# Important Resources

- **Operating Policy Manual**
  - [6.7-1 §4022\(c\) Amounts](#)
  - [6.7-2 DUEC Recovery Amounts](#)
- [Actuarial Technical Manual Section V.J](#)
- [ERISA §4022\(c\)](#)

# Termination Initiation Date (TID)

The TID is defined as:

- Distress Termination –
  - last date on which any Notice of Intent to Terminate (NOIT) is issued to any affected party other than PBGC; or
- PBGC-initiated termination
  - the date the Notice of Determination (NOD) is issued
  - even if there is a NOIT

# SPARR

- The amount paid depends on the Plan's UNGB:
  - Use SPARR
    - $TID \geq 12/18/1990$   
and
      - $UNGB \leq \$20 \text{ Million}$
  - Use Actual Recovery
    - $12/17/1987 < TID < 12/18/1990$   
or
      - $UNGB > \$20 \text{ Million}$

# SPARR

- Small Plan Average Recovery Ratio (SPARR)
  - (Rolling) Average of PBGC's recoveries for all plans over a five-year period
- Pension Protection Act of 1987 (PPA 1987)
  - Five-year period = five fiscal years (10/01-9/30) before the fiscal year in which TID occurs
- Pension Protection Act of 2006 (PPA 2006)
  - Five-year period = five fiscal years ending with the third fiscal year before the fiscal year in which TID occurred

# Small Plan Average Recovery Ratio - SPARR

| SPARR Year | TID on<br>or after | Through    | SPARR (%) |
|------------|--------------------|------------|-----------|
| FY 2018    | 10/01/2017         | 09/30/2018 | 6.84      |
| FY 2017    | 10/01/2016         | 09/30/2017 | 6.92      |
| FY 2016    | 10/01/2015         | 09/30/2016 | 9.39      |
| FY 2015    | 10/01/2014         | 09/30/2015 | 9.29      |
| FY 2014    | 10/01/2013         | 09/30/2014 | 4.93      |

## [Appendix H1 Historical SPARR Information](#)

# Definition

- **Unfunded NonGuaranteed Benefits ("UNGB"):**
  - Total Plan Liabilities – Total Title IV Liabilities



# Determining the Plan's §4022(c) Amount

- §4022(c) Amount = (a) x (b) where
  - (a) UNGB for the plan
  - (b) §4062(b) recovery ratio

# §4022(c) Recovery Ratio

The section §4062(b) recovery ratio for a

- **Non-SPARR Plan:** (UNGBs > 20 Million)
  - PBGC's actual recoveries on its employer liability claims for UBL for that plan

Divided by

- Total UBL for that plan

# §4022(c) Recovery Ratio

The section §4062(b) recovery ratio for a

- **SPARR Plan:** (UNGBs  $\leq$  20 Million)

SPARR = Recovery Ratio

# Allocation Process

- Allocate the plan §4022(c) amount to the plan's UNGB in accordance with ERISA section §4044(a)
- If all UNGB's are satisfied in the higher priority category (lower number), the additional amount is allocated to the next lower priority category (higher number)
- If the §4022(c) amount does not cover the total UNGB for that category or subcategory, the amount is prorated based on Present Values

# Individual §4022(c) Amounts

- **Estimated Benefits**

- PBGC does not normally pay a participant or beneficiary an estimated §4022(c) benefit.

[Policy 6.7-1\(G\)\(1\)](#)

- **Final Benefits**

- PBGC pays final §4022(c) benefits at the same time as PBGC pays final Title IV benefits.

# Form of the §4022(c) Benefit

## Non de Minimis Termination Benefit

- PBGC will pay the individual's §4022(c) amount in the form of the underlying benefit if the benefit is nonforfeitable as of DOPT or is paid as a temporary supplement
- Example: §4022(c)-funded portion of a \$400 temporary supplement (\$20) is paid according to the provisions in the plan (e.g. to age 62 as a SLA). The dollar amount of the supplement will reflect the funded amount.

# Form of the §4022(c) Benefit

## Non de Minimis Termination Benefit

- PBGC will pay the individual's §4022(c) amount as an annuity if the underlying form *is not* a nonforfeitable benefit. The annuity will increase the Title IV benefit.
- Ex: §4022(c) funded portion of a Post-Retirement Death Benefit of \$1000 is converted to an annuity in the same form as the Title IV Benefit and that amount added to the monthly annuity.

# Form of the §4022(c) Benefit

## Non de Minimis Termination Benefit

- The additional *not nonforfeitable* §4022(c) monthly benefit (previous slide) is calculated as of XRD in the XRD valuation form.
- For recalculation purposes, the additional *not nonforfeitable* §4022(c) monthly benefit is divided by a ratio of ERFs and BFCFs to move the amount to the participant's actual ARD



# Example 1

## Plan Information

- PV of Title IV Benefits: \$3,268,277
- Plan provides a pre-retirement Lump Sum Death Benefit
  - Not a nonforfeitable benefit
  - Entire value of Benefit is in PC6
- No other non-guaranteed benefits
- Entire §4022(c) amount allocated to PC6
- PV of PC6 for all participants = \$4,389

# Example 1

## Plan Information

- No §4044 asset allocation required
  - GMB = Plan Benefits except Pre-Ret LS DB
  - Assets do not cover Title IV Benefits; no §4044 asset allocation in PC6

# Example 1

- $UNGB = PV\ PC6 = \$4,389$
- Plan §4022(c) Amount:
  - $PVPC6(UNGB) \times FY14\ SPARR$
  - $\$4,389 \times 4.93\% = \$215$  (\$216 - \$1 rounding)
- §4022(c) Funds 4.93% of PC6

# Example 1

## Participant “D”

- Title IV monthly benefit = \$385.83 (at XRD in the XRD form)
- PV of Death Benefit in PC6 = \$709
- P’s §4022(c) amount:  $\$709 \times .0493 = \$35$
- P’s §4022(c) Benefit: \$35 converted to annuity = \$0.20 (at XRD in the XRD form)
- P’s Termination Benefit:  $\$385.83 + \$0.20 = \$386.03$

# Example 2

## Plan Information

### Pension Plan for Pete's Professional Car Wash

- DOPT: 06/15/2006
- NOD: 02/19/2008
- NOIT: 4/15/2006
- DOTR: 7/18/2008
- Original Plan Effective Date: 1/01/1980

# Example 2

## Pension Plan for Pete's Professional Car Wash

### Plan Provisions

- NRA: 65
- ERA: 55
- EURA: 62
- Automatic Single Form: SLA
- Automatic Married Form: J&50%S
- Expense Load: 1.5097%

# Example 2

## Pension Plan for Pete's Professional Car Wash

### Plan Provisions

- $NRB = [.015 \times AMC + .005 \times (AMC - \$800)] \times YOP$
- Vesting Schedule: 5 Year Cliff
- ERF: Plan Provided Table
- FCF: UP1984, IR=7.00%

## Example 2

# Pension Plan for Pete's Professional Car Wash

- Present Value of Title IV Benefits: \$10,389,063
- Assets Available for Allocation: \$7,035,429
- Present Value of all Benefits (BL): \$10,474,461
- Present Value of PC3: \$5,434,129
- Assets cover 100% of PC3
- Remaining assets available to allocate to PC4:
  - $\$7,035,429 - \$5,434,129 = \$1,601,300$



## Example 2

# Pension Plan for Pete's Professional Car Wash

- Assets cover 32.3173% of PC4 Benefits
  - $\$1,601,300 / (\$10,389,063 - \$5,434,129) = 32.3173\%$
- $UBL = BL - \text{Plan's Valuation Assets}$ 
  - $\$10,474,461 - \$7,035,429 = \$3,439,032$
- $UNGB < \$20M \rightarrow \text{SPARR Plan}$
- SPARR: 4.26%
  - Based on TID = NOD in FY08

# Example 2

## Pension Plan for Pete's Professional Car Wash

- Participant Pete
  - Status: Active at DOPT
  - DOB: 07/09/1948
  - ANB: 58
  - Marital Status: Unknown
  - Estimated SDOB:07/09/1952
  - Majority Owner

# Example 2

- Participant Pete
  - XRD: 8/01/2008 (AA60)
  - Form of Annuity for valuation: J&50%S
  - NRB: \$3,612.50
  - Plan Benefit calculated at XRD:
    - $\$3,612.50 \times .9042 \times .8670 = \$2,831.99$
    - QPSA Benefit =  $\$2,831.99 \times 50\% = \$1,416.00$
  - PC3 Benefit: \$2,156.09
    - QPSA Benefit =  $\$2,156.09 \times 50\% = \$1,078.05$

# Example 2

- Participant Pete
  - No Majority Owner phase-in since plan >10 years old
  - MGB at XRD:
    - $\$3,971.59 \times .6558 \times .90 \times .96 = \$2,250.35$
  - [ATM IV.A.2, MGB Tables, Reduction Factors for MGB](#)
  - Benefit in excess of the MGB:
    - $\$2,831.99 - \$2,250.35 = \$581.64$

# Example 2

- Participant Pete
  - Plan Benefit: \$2,831.99
  - GMB = \$2,250.35 (Equals MGB)
  - PC3 Funded Benefit: \$2,156.09
  - No additional benefit provided by a \$4044 allocation in PC3 since  $PC3 < GMB$
  - Net PC4 Benefit =  $GMB - PC3 \text{ Benefit} = \$94.26$
  - Net PC5 Benefit: Plan Benefit – GMB
    - $\$2,831.99 - \$2,250.35 = \$581.64$

# Present Value Factors

- Participant Pete
  - ANB = 58 deferred to XRA = 60
  - J&50%S; Spouse assumed to be 4 years younger
  - Male
  - PVF = 11.5548
- QPSA Factor
  - Probability of p's death between 58 and 60
  - SANB = 54 deferred to SXRA = 56
  - SLA; Female
  - QPSA PVF = 0.1210

# Example 2

- Participant Pete
  - **PV of Plan Benefit:**
    - $\$2,831.99 \times 12 \times 11.5548$  ( \$392,677)  
+  $\$1,416.00 \times 12 \times .1210$  (\$2,056) = \$394,733
  - Expense Load:  $\$394,733 \times 1.015097 = \mathbf{\$400,692}$
  - **PV of PC3 Benefit:**
    - $\$2,156.09 \times 12 \times 11.5548$  (\$298,958)  
+  $\$1,078.05 \times 12 \times .1210$  (\$1,565) = \$300,523
  - Expense Load:  $\$300,523 \times 1.015097 = \mathbf{\$305,060}$

# Example 2

- Participant Pete
  - **PV of Net PC4 Benefit:**
    - $\$94.26 \times 12 \times 11.5548$  (\$13,070)  
+  $\$47.13 \times 12 \times .1210$  (\$68) = \$13,138
  - Expense Load:  $\$13,138 \times 1.015097 = \$13,336$
  - **PV of Net PC5 Benefit:**
    - $\$581.64 \times 12 \times 11.5548$  (\$80,649)  
+  $\$290.82 \times 12 \times .1210$  (\$422) = \$81,071
  - Expense Load:  $\$81,071 \times 1.015097 = \$82,295$



# PV by PC for Pete

## before §4022(c) Allocation

| PC    | Benefit    | PV of Benefit | QPSA Benefit | PV of QPSA | TOTAL PV w/Load |
|-------|------------|---------------|--------------|------------|-----------------|
| 1     | \$0        | \$0           | \$0          | \$0        | \$0             |
| 2     | \$0        | \$0           | \$0          | \$0        | \$0             |
| 3     | \$2,156.09 | \$298,958     | 1,078.05     | \$1,565    | \$305,060       |
| 4     | \$94.26    | \$13,070      | \$47.13      | \$68       | \$13,336        |
| 5     | \$581.64   | \$80,649      | \$290.82     | \$422      | \$82,295        |
| 6     | \$0        | \$0           | \$0          | \$0        | \$0             |
| TOTAL | \$2,831.99 | \$392,677     | \$1,416.00   | \$2,055    | \$400,692       |

# Example 2

## Non-Vested Participants

- Nick
  - Accrued Benefit (AMB): \$77.05
  - PV\* of XRB: \$2,359
- Nan
  - Accrued Benefit (AMB): \$33.73
  - PV\* of XRB: \$744
- Total PV of PC6:  $\$2,359 + \$744 = \$3,103$

\* PV includes QPSA and expense load

# Plan PV by PC

## before §4022(c)Allocation

| Status   | PC3         | PC4         | PC5      | PC6     | BL           |
|----------|-------------|-------------|----------|---------|--------------|
| Retirees | \$3,246,849 | \$372,697   | \$0      | \$0     | \$3,619,546  |
| SV       | \$987,456   | \$2,217,709 | \$0      | \$0     | \$3,205,165  |
| AV       | \$1,199,824 | \$2,364,528 | \$82,295 | \$0     | \$3,646,647  |
| NV       | \$0         | \$0         | \$0      | \$3,103 | \$3,103      |
| TOTAL    | \$5,434,129 | \$4,954,934 | \$82,295 | \$3,103 | \$10,474,461 |

# §4022 (c) Amount

- §4022(c) Amount = (a) x (b) where
  - (a) UNGB for the plan
  - (b) §4062(b) recovery ratio
- Valuation UNGB:  $\$82,295 + \$3,103 = \$85,398$
- SPARR = 4.26%
- Plan's §4022(c) Amount:
  - $\$85,398 \times .0426 = \$3,638$

# Plan's §4022(c) Amount

- Allocate \$3,638 by Priority Category
- Entire amount allocated to PC5
- Non-Vested participant receive \$0
- PV of Net PC5 = \$82,295
- Allocation Ratio =  $\$ 3,638 / \$ 82,295 = .044207$

# Pete's §4022(c) Amount

- PV of Pete's PC5 = \$82,295
- Amount funded through §4022(c)
  - \$ 82,295 x 0.044207 = \$3,638
- Benefit provided to participant
  - \$581.64 x 0.044207 = \$25.71

# Pete's Termination Benefit

- Title IV Benefit = GMB + funded §4044 Benefit
  - $= \$2,250.35 + \$0 = \$2,250.35$
- Termination Benefit:
  - Title IV Benefit + §4022(c) Benefit
  - $\$2,250.35 + \$25.71 = \$2,276.06$

# PV of Pete's Termination Benefit

- PV of Termination Benefit:
- PV of the Title IV Benefit + §4022(c) amount:
  - $\$318,396 + \$3,638^* = \$322,034$

\* Use actual amount of PV rounded to the nearest dollar for each participant



# Final Plan Valuation Liability

| Group    | PV Title IV  | PV UNGB  | PV Benefit Liabilities | PV \$4022(c) Benefits | PV Termination Benefits |
|----------|--------------|----------|------------------------|-----------------------|-------------------------|
| Retirees | \$3,619,546  | \$0      | \$3,619,546            | \$0                   | \$3,619,546             |
| SV       | \$3,205,165  | \$0      | \$3,205,165            | \$0                   | \$3,205,165             |
| AV       | \$3,564,352  | \$82,295 | \$3,646,647            | \$3,638               | \$3,567,990             |
| NV       | \$0          | \$3,103  | \$3,103                | \$0                   | \$0                     |
| TOTAL    | \$10,389,063 | \$85,398 | \$10,474,461           | \$3,638               | \$10,392,701            |

# In Class Assignment

1. Calculate Pete's Termination Benefit at
  - EURD: 8/01/2010
  - PBGC's ERF (AA62) = .790
  - Straight Life Annuity
2. Calculate the §4022(c) Benefits assuming the assets are \$10,471,358

# Answer

1. Pete's Benefit commencing 8/01/2010:

a) Maximum Guaranteeable Benefit:

- $\$3,971.59 \times .790 = \$3,137.56$

b) Benefit in excess of the MGB:

- $\$3,612.50 \text{ (EURB)} - \$3,137.56 = \$474.94$

c) §4022(c) Benefit:

- $\$474.94 \times 0.044207 = \$21.00$

d) PBGC Termination Benefit

- $\$3,137.56 + \$21.00 = \$3,158.56$

# Answers

2. Assets = \$10,471,358
- a) The assets cover the benefits through PC5
  - b) §4022(c) amount goes to PC6
  - c) UNGB in PC6 = \$3,103
  - d) SPARR = .0426 (4.26%)
  - e) §4022(c) amount = \$3,103 x .0426 = \$132
  - f) Split between Nick and Nan based on PV

# Answers

2. Assets = \$10,471,358

g. Nick:  $PV = \$2,359 \times .0426 = \$100$

h. Nan:  $PV = \$744 \times .0426 = \$32$

Since the  $PV < \$5,000$  for each benefit, Nick and Nan will each receive a lump sum payment. The amount above must be converted to an annuity and then revalued without the QPSA using the LS rates.

# Questions

07/16/2026

*Office of Benefits Administration (OBA)*

